

1 Write a Python program using Pandas to read a CSV file and display its contents.

Aim

To write a Python program that uses the Pandas library to read a CSV file and display its contents in tabular form.

Algorithm

1. Import the library.
2. Use to read the CSV file.
3. Store the data in a DataFrame.
4. Display the DataFrame contents using or directly calling the DataFrame.
5. Optionally, use methods like or to view specific rows.

CODE

```
import pandas as pd
```

```
# Step 1: Read CSV file
```

```
# Replace 'data.csv' with the actual filename
```

```
df = pd.read_csv("data.csv")
```

```
# Step 2: Display contents
```

```
print("Contents of the CSV file:\n")
```

```
print(df)
```

```
# Optional: Display first 5 rows
```

```
print("\nFirst 5 rows of the CSV file:")
```

```
print(df.head())
```

```
# Optional: Display last 5 rows
```

```
print("\nLast 5 rows of the CSV file:")  
print(df.tail())
```

OUTPUT

Contents of the CSV file:

	Name	Age	Marks
0	Yogin	20	89
1	Priya	21	95
2	Alex	19	78
3	Sam	22	88

Conclusion

- The program uses Pandas to read and display CSV data.
- Pandas makes handling tabular data simple and efficient.
- Methods like `head()` and `tail()` allow quick inspection of large datasets.

2 Write a program to create a Series and perform operations on it.

Aim

To write a Python program that creates a Pandas Series and performs basic operations like indexing, slicing, mathematical operations, and statistical functions.

Algorithm

1. Import the `pandas` library.
2. Create a Series from a list of numbers.
3. Perform operations such as:
 - Accessing elements by index.
 - Slicing a range of elements.

- Applying mathematical operations (e.g., multiply by 2).
 - Using statistical functions (, , ,).
4. Display results.

CODE

```
import pandas as pd
```

```
# Step 1: Create a Series
```

```
data = [10, 20, 30, 40, 50]
```

```
series = pd.Series(data)
```

```
print("Original Series:")
```

```
print(series)
```

```
# Step 2: Access elements
```

```
print("\nAccess element at index 2:", series[2])
```

```
# Step 3: Slicing
```

```
print("\nSlicing from index 1 to 3:")
```

```
print(series[1:4])
```

```
# Step 4: Mathematical operation
```

```
print("\nMultiply each element by 2:")
```

```
print(series * 2)
```

```
# Step 5: Statistical operations
```

```
print("\nSum of elements:", series.sum())
```

```
print("Mean of elements:", series.mean())
```

```
print("Maximum value:", series.max())
```

```
print("Minimum value:", series.min())
```

OUTPUT

Original Series:

0 10

1 20

2 30

3 40

4 50

dtype: int64

Access element at index 2: 30

Slicing from index 1 to 3:

1 20

2 30

3 40

dtype: int64

Multiply each element by 2:

0 20

1 40

2 60

3 80

4 100

dtype: int64

Sum of elements: 150

Mean of elements: 30.0

Maximum value: 50

Minimum value: 10

Conclusion

- The program demonstrates how to create a Pandas Series.
- It shows indexing, slicing, mathematical operations, and statistical functions.
- Pandas makes handling numerical data simple and efficient